```c
#include <stdio.h>
#include <string.h>

#define NUM_STATES 8
#define NUM_INPUTS 4

// DFA States
enum { D0, D1, D2, D3, D4, D5, D6, DEAD };

// input mapper
int get_input(char c) {
   if (c=='a') return 0;
   if (c=='b') return 1;
   if (c=='c') return 2;
   return 3;
}

// accepting tokens
const char* accepting_tokens[NUM_STATES] = {
   NULL,      // D0
   NULL,      // D1
   NULL,      // D2
   NULL,      // D3
   "aa*b",    // D4
   "bb*c",    // D5
   "cc*a",    // D6
   NULL       // DEAD
};

int main() {

   int next_state[NUM_STATES][NUM_INPUTS] = {

     /* D0 */ { D1, D2, D3, DEAD },
```

```
    /* D1 */ { D1, D4, DEAD, DEAD },
    /* D2 */ { DEAD, D2, D5, DEAD },
    /* D3 */ { D6, DEAD, D3, DEAD },
    /* D4 */ { DEAD, DEAD, DEAD, DEAD },
    /* D5 */ { DEAD, DEAD, DEAD, DEAD },
    /* D6 */ { DEAD, DEAD, DEAD, DEAD },
    /* DEAD */{ DEAD, DEAD, DEAD, DEAD }
  };

  char str[100];
  printf("Enter string: ");
  fgets(str,sizeof(str),stdin);
  str[strcspn(str,"\n")] = 0;

  int state = D0;

  for(int i=0; str[i]!='\0'; i++){
    int input = get_input(str[i]);
    state = next_state[state][input];
    if(state==DEAD) break;
  }

  if(accepting_tokens[state])
    printf("%s Accepted (%s)\n",str,accepting_tokens[state]);
  else
    printf("%s Rejected\n",str);

  return 0;
}
```

```
nts/GitHub/compiler-design/labReport1Code/"1example
Enter string: aaab
aaab Accepted (aa*b)

nts/GitHub/compiler-design/labReport1Code/"1example
Enter string: bbbbbbbc
bbbbbbbc Accepted (bb*c)

nts/GitHub/compiler-design/labReport1Code/"1example
Enter string: cccccccccac
cccccccccac Rejected
```

```c
#include <stdio.h>
#include <string.h>

#define NUM_STATES 8
#define NUM_INPUTS 4

enum { D0, D1, D2, D3, D4, D5, D6, DEAD };

int get_input(char c){
    if(c=='a') return 0;
    if(c=='b') return 1;
    if(c=='c') return 2;
    return 3;
}

const char* accepting_tokens[NUM_STATES] = {
    NULL,      // D0
    NULL,      // D1
    NULL,      // D2
    "b*c",     // D3
    "a+b",     // D4
    "b*c",     // D5
    "ca+",     // D6
    NULL       // DEAD
};

int main(){

    int next_state[NUM_STATES][NUM_INPUTS] = {

      /* D0 */ { D1, D2, D3, DEAD },
      /* D1 */ { D1, D4, DEAD, DEAD },
      /* D2 */ { DEAD, D2, D5, DEAD },
```

```
    /* D3 */ { D6, DEAD, DEAD, DEAD },
    /* D4 */ { DEAD, DEAD, DEAD, DEAD },
    /* D5 */ { DEAD, DEAD, DEAD, DEAD },
    /* D6 */ { D6, DEAD, DEAD, DEAD },
    /* DEAD */{ DEAD, DEAD, DEAD, DEAD }
  };

  char str[100];
  printf("Enter string: ");
  fgets(str,sizeof(str),stdin);
  str[strcspn(str,"\n")] = 0;

  int state = D0;

  for(int i=0; str[i]!='\0'; i++){
      int input = get_input(str[i]);
      state = next_state[state][input];
      if(state==DEAD) break;
  }

  if(accepting_tokens[state])
      printf("%s Accepted (%s)\n",str,accepting_tokens[state]);
  else
      printf("%s Rejected\n",str);

  return 0;
}
```

```
ents/GitHub/compiler-design/labReport1Code/"2example
Enter string: 12s
12s Rejected

nts/GitHub/compiler-design/labReport1Code/"2example
Enter string: caaa
caaa Accepted (ca+)

nts/GitHub/compiler-design/labReport1Code/"2example
Enter string: aab
aab Accepted (a+b)
```